

Born and Inflationary Probabilities as a Common-Measure

Obligation:

Exact Pushforward Criteria and a Conditional MTT Model

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Abstract

The quantum Born rule and the inflationary measure problem both involve probability after reduction to an effective description, but this resemblance does not prove a common physical origin. We formulate the proposed relation as a typed measure-theoretic obligation. A successful bridge must supply one selected source space and probability law or probability kernel, separate quantum and cosmological observable maps, and verified pushforward equalities on declared domains. Two elementary results expose the difficulty. A non-injective projection does not create probability, and any two prescribed target laws admit a trivial common realization by a product measure. Common origin becomes explanatory only when the source and maps are fixed before the target laws enter. The current MTT q79 construction supplies a selected quantum leg for its canonical binary one-anchor recorder, but not arbitrary apparatus contexts; no selected cosmological history measure or same-source bridge is yet available. We retain the earlier e-fold weighting as a conditional model. For positive parameters, its generalized density is normalizable, has a unique mode, and has an exact curvature pushforward, but its front, barrier, and volume parameters remain ansatz data rather than MTT predictions. The result is a precise common-measure research program, not a derivation of the Born rule, inflationary model preference, or a universal cosmological measure.

Version 2 Revision Note

Supersedes *Inflationary Measures and the Born Rule as a Single Shadow–Bridge Problem*, version 1.

Reason The earlier paper treated a schematic basin functional as selected, inferred the Born rule from normalized basin weights, and described an ansatz e-fold density as a regulator-free MTT prediction. It did not specify the source probability law, context dependence, cosmological quotient, or typed pushforward maps, and it used ordinary observational model preferences as validation of the proposed weighting.

Resolution This version gives an exact common-measure contract, proves that non-injective projection alone does not generate probabilities, and proves that unconstrained common-source realizations are vacuous. It records the current restricted q79 Born result, isolates the open cosmological and same-source gates, and reclassifies the e-fold density as a conditional four-parameter model whose normalization, mode, and curvature pushforward are proved explicitly.

Retained result Born and cosmological probabilities can meaningfully be compared as typed pushforwards, and the displayed admissibility-weighted e-fold family is a mathematically well-defined illustrative model.

Remaining boundary MTT has not selected one normalized upper measure with both the required general Born pushforwards and a gauge-invariant cosmological-history pushforward. The e-fold front and weight parameters are not derived from q79 geometry or its selected action.

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1 One analogy, three different questions

The Born rule and the cosmological measure problem are often placed under the same heading because both ask for probabilities. Their mathematical jobs are different.

Quantum outcome law Given a preparation, instrument, and recorded outcome algebra, determine the probabilities of the possible records.

Cosmological history law Given dynamics, initial or boundary data, and an observational conditioning rule, assign weights to histories or events without an arbitrary regulator or slicing dependence.

Common-source claim Show that both laws are pushforwards of one independently selected upper probability construction.

The third claim is much stronger than the first two. A derivation of one quantum recorder does not define a measure on inflationary histories. Conversely, a cosmological cutoff prescription does not imply the quadratic quantum law.

The useful MTT intuition is that both descriptions may discard upper information. This motivates a common diagram. It does not fill the diagram. The source measure, the observable maps, their domains, and the relevant conditioning must all be supplied.

2 The typed common-measure contract

2.1 Fixed-context form

Let

$$(\mathcal{U}, \Sigma_{\mathcal{U}})$$

be a source measurable space and let μ be a probability measure on it. Let

$$(\Omega_Q, \Sigma_Q), \quad (\Omega_C, \Sigma_C)$$

be a quantum outcome space and a cosmological observable space. Measurable maps

$$\pi_Q : \mathcal{U} \longrightarrow \Omega_Q, \quad \pi_C : \mathcal{U} \longrightarrow \Omega_C$$

induce the pushforward laws

$$\nu_Q = (\pi_Q)_* \mu, \quad \nu_C = (\pi_C)_* \mu, \quad (2.1)$$

where

$$(\pi_Q)_* \mu(E) = \mu(\pi_Q^{-1} E)$$

and similarly for π_C .

Definition 2.1 (Strong fixed-context bridge). A strong common-measure bridge consists of

$$(\mathcal{U}, \Sigma_{\mathcal{U}}, \mu, \pi_Q, \pi_C)$$

together with proofs that:

1. μ is normalized and selected without using either target law as construction data;
2. both maps are defined on the same physical source, not merely on isomorphic copies;
3. the first equality in (2.1) is the intended quantum outcome law;
4. the second equality is well defined on gauge-equivalence classes of cosmological histories and is independent of arbitrary regulator choices; and
5. all conditioning, null sets, domains, and approximation errors are stated.

2.2 Contextual form

One fixed measure cannot express all preparations and instruments unless those choices are included in the source or in its conditioning. Let $\mathcal{C}$ be a context space. The appropriate source object is generally a probability kernel

$$\mathbf{M} : \mathcal{C} \times \Sigma_{\mathcal{U}} \longrightarrow [0, 1], \quad c \longmapsto \mu_c.$$

Quantum contexts may include a preparation, a measurement instrument, and a stopping rule. Cosmological contexts may include a background branch, conditioning data, and an observer/event prescription. A full bridge must explain which context variables are physical inputs and which are selected by the theory. Hiding all context dependence in the symbol μ only disguises the parameter ledger.

3 Two exact guardrails

3.1 Non-injectivity does not create probability

Proposition 3.1 (Projection guardrail). *Let $\pi : \mathcal{U} \rightarrow \Omega$ be any measurable map, including a non-injective one. The map alone does not determine a probability law on Ω . In particular, if the upper state is the point measure δ_u , then*

$$\pi_*\delta_u = \delta_{\pi(u)}.$$

Proof. For every measurable $E \subseteq \Omega$,

$$(\pi_*\delta_u)(E) = \delta_u(\pi^{-1}E) = \begin{cases} 1, & \pi(u) \in E, \\ 0, & \pi(u) \notin E. \end{cases}$$

This is exactly $\delta_{\pi(u)}$. Without an upper measure or state there is no object to push forward. $\square$

Many-to-one projection explains loss of reconstructibility: distinct upper points may have the same retained value. It does not choose how much weight to place on those points. That weight must come from a state, invariant measure, normal functional, ensemble, or dynamical sampling theorem.

3.2 An unconstrained common source is automatic

Proposition 3.2 (Trivial common realization). *Let ν_Q and ν_C be arbitrary probability measures on (Ω_Q, Σ_Q) and (Ω_C, Σ_C) . Then there exists a common source measure with those two pushforwards.*

Proof. Take

$$\mathcal{U} = \Omega_Q \times \Omega_C, \quad \mu = \nu_Q \otimes \nu_C,$$

and let π_Q, π_C be the coordinate projections. The defining property of the product measure gives

$$(\pi_Q)_*\mu = \nu_Q, \quad (\pi_C)_*\mu = \nu_C.$$

$\square$

Corollary 3.3 (Non-explanatory existence). *The bare statement that the Born law and a cosmological law are projections of one measure has no predictive content. The scientific content must lie in the prior selection of $\mathcal{U}$, μ , the two maps, and their context dependence.*

The same conclusion holds with correlated couplings: choosing a joint law with the desired marginals can encode any permitted correlation. A common source becomes evidence only when it is independently constrained enough that the target laws could have come out differently.

4 Current MTT status

4.1 What the q79 quantum construction supplies

The current MTT ledger contains a stronger quantum result than was available to version 1. On the canonical q79 binary one-anchor recorder, the selected normal state on the commuting

Fock output algebra emits the stopped output measure, and the second-moment capture descent is exact on that domain. No observed probability is inserted as a fit coordinate.

This is a genuine selected quantum leg, but its quantifiers matter. It does not yet cover every allowed apparatus context, finite-bandwidth or non-Markov control, or an objective law selecting one ontic history. The general Born source theorem therefore remains open even though the canonical recorder is closed.

4.2 What is not yet supplied

Required object	Current status
Selected q79 source for canonical binary recorder	Available on the declared one-anchor domain
Quantum output pushforward	Exact on that domain
Arbitrary physical instruments and controls	Open
Objective one-history selection	Open
Selected inflationary history space	Open
Gauge- and slicing-invariant history sigma-algebra	Open
Cosmological source law	Open
Cosmological readout map	Open
Same physical source for both legs	Open

Accordingly, the current status of the common Born–cosmology bridge is interpretive. The exact quantum leg narrows the missing work; it does not promote the cosmological leg.

5 A conditional e-fold model

The earlier paper proposed a coherence front N_c , a linear margin

$$C(N) = \kappa(N_c - N),$$

a barrier $\Delta A/\hbar = \lambda/C(N)^2$, and physical-volume weighting e^{3N} . Those choices are not consequences of projection. They define a model. It is useful to analyze the whole family transparently.

5.1 Definition and normalization

For

$$\alpha > 0, \quad \beta > 0, \quad p > 0, \quad N_c \in \mathbb{R},$$

define on $N < N_c$

$$w_{\alpha,\beta,p,N_c}(N) = \exp\left(\beta N - \frac{\alpha}{(N_c - N)^p}\right). \quad (5.1)$$

The original ansatz is recovered with

$$\beta = 3, \quad p = 2, \quad \alpha = \frac{\lambda}{\kappa^2}.$$

Theorem 5.1 (Conditional e-fold density). *The integral*

$$Z_{\alpha,\beta,p,N_c} = \int_{-\infty}^{N_c} w_{\alpha,\beta,p,N_c}(N) dN$$

is finite and positive. The normalized density

$$f_N(N) = Z_{\alpha,\beta,p,N_c}^{-1} w_{\alpha,\beta,p,N_c}(N)$$

has a unique mode at

$$N_{\text{mode}} = N_c - \left(\frac{\alpha p}{\beta} \right)^{1/(p+1)}. \quad (5.2)$$

All polynomial moments of $N_c - N$ are finite.

Proof. Set $x = N_c - N > 0$. Then

$$Z_{\alpha,\beta,p,N_c} = e^{\beta N_c} \int_0^\infty \exp\left(-\beta x - \frac{\alpha}{x^p}\right) dx.$$

For $x \geq 1$, the integrand is bounded by $e^{-\beta x}$. For $0 < x \leq 1$, it is bounded by $e^{-\alpha/x^p}$, whose integral is finite after the substitution $y = \alpha/x^p$. The same estimates with an extra factor x^m prove finiteness of every nonnegative polynomial moment.

The log density in the x coordinate is

$$\ell(x) = -\beta x - \frac{\alpha}{x^p}.$$

Its derivative vanishes only when

$$-\beta + \frac{\alpha p}{x^{p+1}} = 0,$$

giving $x_{\text{mode}} = (\alpha p / \beta)^{1/(p+1)}$. Moreover,

$$\ell''(x) = -\frac{\alpha p(p+1)}{x^{p+2}} < 0,$$

so this stationary point is the unique mode. Returning to $N = N_c - x$ gives (5.2). $\square$

5.2 What each parameter means

Symbol	Model role	Source status
N_c	location of the proposed admissibility front	not selected by current MTT geometry
β	exponential history/volume weight	$\beta = 3$ assumes three-dimensional physical-volume weighting
p	barrier exponent	$p = 2$ is an ansatz, not a theorem
α	barrier scale	combines the proposed stiffness and margin normalization

Normalization removes one overall multiplicative constant, but it does not remove these four shape and location inputs. A fitted or hand-selected member of this family is therefore not a zero-parameter prediction.

6 Exact curvature pushforward

Suppose, only for this model, that a positive curvature proxy obeys

$$\Omega = \Omega_\star e^{-2N}, \quad \Omega_\star > 0. \quad (6.1)$$

The map is monotone and has inverse

$$N(\Omega) = -\frac{1}{2} \log\left(\frac{\Omega}{\Omega_\star}\right), \quad \left|\frac{dN}{d\Omega}\right| = \frac{1}{2\Omega}.$$

Writing

$$\Omega_c = \Omega_\star e^{-2N_c},$$

the exact pushforward density on $\Omega > \Omega_c$ is

$$f_\Omega(\Omega) = \frac{1}{2\Omega Z_{\alpha,\beta,p,N_c}} \exp\left[-\frac{\beta}{2} \log\left(\frac{\Omega}{\Omega_\star}\right) - \frac{\alpha}{\left(N_c + \frac{1}{2} \log(\Omega/\Omega_\star)\right)^p}\right]. \quad (6.2)$$

Equation (6.2) is an exact change of variables. It does not establish that (6.1) is valid across all histories, fix $\Omega_\star$, supply a reheating map, or select the four e-fold parameters. In particular, normalizability of a chosen density is not a solution of the eternal-inflation measure problem. The latter also requires a physical event algebra, regulator independence, conditioning, and a derivation from the underlying dynamics [1].

7 What the observational comparison does and does not test

At leading slow-roll order, representative model curves include

$$\text{Starobinsky-like:} \quad n_s \simeq 1 - \frac{2}{N}, \quad r \simeq \frac{12}{N^2}, \quad (7.1)$$

$$V(\phi) \propto \phi^m : \quad n_s \simeq 1 - \frac{m+2}{2N}, \quad r \simeq \frac{4m}{N}. \quad (7.2)$$

For $N = 50\text{--}60$, (7.1) gives

$$n_s \simeq 0.960\text{--}0.967, \quad r \simeq 0.0048\text{--}0.0033.$$

The minimally coupled quadratic model gives

$$r \simeq 0.160\text{--}0.133,$$

and the quartic model gives

$$r \simeq 0.320\text{--}0.267.$$

For historical, reproducible reference, Planck 2018 reported $n_s = 0.9649 \pm 0.0042$ at 68% confidence and, with BK15, $r_{0.002} < 0.056$ at 95% confidence [2]. The later BICEP/Keck analysis through the 2018 observing season reported $r_{0.05} < 0.036$ at 95% confidence [3].

These comparisons show that the simple monomial examples have much larger tensor amplitudes than the plateau example. They do not validate (5.1). The model curves already have those properties before any admissibility weighting is applied. A weight over N can change how points along a curve are averaged; it cannot turn ordinary agreement of a model curve with data into evidence for the proposed upper measure.

A genuine test would preselect the e-fold density and reheating map, propagate them through a declared likelihood, and compare held-out observables against specified alternatives. No such model-selection pipeline is claimed here.

8 The exact completion target

The common-measure proposal becomes a physical MTT theorem only after the following objects are constructed from one branch.

1. **Source.** A selected measurable upper state space and normalized measure or probability kernel, with its dependence on branch and context declared.
2. **Quantum map.** A family of selected record maps covering the intended preparations and instruments, together with the Born pushforward equalities and error bounds.
3. **Cosmological quotient.** A selected history space, its gauge-equivalence relation, and a sigma-algebra of physical events on the quotient.
4. **Cosmological map.** A measurable map to late-time observables or event classes whose pushforward is finite and independent of arbitrary time slicing or cutoff choices.
5. **Same-source certificate.** A commuting diagram showing that both maps originate from the same selected upper state and action, preserving the relevant connections, conditioning, and branch data.
6. **Non-replay test.** A proof that the construction did not use the target Born or cosmological probabilities as source inputs, followed by at least one held-out prediction or exclusion.

In compressed form, the desired diagram is

$$\begin{array}{ccc}
 & (\mathcal{U}, \mu_c) & \\
 \swarrow \pi_{Q,c} & & \searrow \pi_{C,c} \\
 (\Omega_{Q,c}, \nu_{Q,c}) & & (\Omega_{C,c}, \nu_{C,c}),
 \end{array}$$

with both displayed target measures proved to be pushforwards and with the source selected before either target law is read.

9 Interpretive consequences

The corrected framework preserves a useful conceptual lesson. Quantum records and cosmological observations can both be described by conditioning and pushforward from richer state spaces. Their effective probability problems may therefore share mathematical tools: disintegration, stopping rules, invariant measures, quotient sigma-algebras, and recovery criteria.

Several stronger conclusions do not follow:

- projection by itself does not produce randomness or the Born rule;
- normalizing one e-fold ansatz does not regulate eternal inflation;
- a preference for plateau model curves does not select the ansatz;
- the compact shared MTT circle is not thereby identified with physical time;
- low initial entropy is not derived from “basin entry”;
- quantum measurement is an ordinary physical record-forming process, not a privileged cosmological event; and
- failure of one effective encoding does not prove that the underlying universe has crossed a literal coherence wall.

These boundaries strengthen the research program. They turn an analogy into a finite list of constructions whose success or failure can be checked.

10 Conclusion

The Born rule and inflationary measures can be placed in one exact measure-theoretic language, but they have not yet been derived from one MTT source. Non-injective projection explains why upper distinctions may be lost; it does not supply their weights. Conversely, any desired pair of target laws can be manufactured as marginals of a product measure, so a freely chosen common source proves nothing.

The nontrivial theorem sought by MTT is therefore a selection theorem. One upper state, action, and probability kernel must be fixed independently, then shown to produce the quantum and cosmological laws through typed physical maps. The q79 binary one-anchor recorder now supplies a restricted quantum endpoint. The cosmological history measure and the same-source intertwiner remain open.

The admissibility-weighted e-fold family remains mathematically useful as a conditional model: it is normalizable, unimodal, and has an exact curvature pushforward. Its four continuous inputs are currently assumptions. This is the proper present status of the shadow-bridge idea: a sharpened and testable program, not yet a unified derivation of physical probability.

Computational Evidence and Reproducibility

The common-measure criterion is a measure-theoretic obligation. The internal TT support certificate is relevant geometric context for gravitational sectors, but it does not select either a Born measure or an inflationary measure and is therefore not used as direct proof here.

The referenced rows are frozen to the curated results repository state identified below. Hashes are grouped in eight-character blocks for line breaking.

Repository: <https://github.com/PeterNero/mtt-results-repro>
Commit: 31247ebb 5c22f3fb b5443024 365433c6 ee0bff4a
Manifest: release/result_manifest.json
Manifest SHA-256:
fb399689 60b00584 631dbf53 1a708e18
ef928d6b 6d935119 c185d7f6 32b1e7cd

Tier labels are quoted verbatim from that manifest. A row used directly supports only the specific computational statement identified above; a corpus-state cross-check does not prove this paper's local theorems; and an open row is evidence of an unresolved obligation, never of closure.

Corpus-state cross-checks.

- **A13/gr_tt_support** (*derived exact*).

Exact-branch internal TT support certificate; physical normalization remains open.

No imported row changes theorem ownership or promotes a neighboring claim: all local statements retain their stated hypotheses, domains, and limitations.

References

- [1] B. Freivogel, *Making predictions in the multiverse*, Classical and Quantum Gravity **28** (2011) 204007, doi:10.1088/0264-9381/28/20/204007, arXiv:1105.0244.
- [2] Y. Akrami et al. (Planck Collaboration), *Planck 2018 results. X. Constraints on inflation*, Astronomy & Astrophysics **641** (2020) A10, doi:10.1051/0004-6361/201833887, arXiv:1807.06211.
- [3] P. A. R. Ade et al. (BICEP/Keck Collaboration), *BICEP/Keck XIII: Improved Constraints on Primordial Gravitational Waves using Planck, WMAP, and BICEP/Keck Observations through the 2018 Observing Season*, Physical Review Letters **127** (2021) 151301, doi:10.1103/PhysRevLett.127.151301, arXiv:2110.00483.